

Data sets required for the S.Europe Climate Change Project :

Here's the exact dataset list for your narrowed plan (Seville + Larissa, 1990–2020 baseline, SSP5-8.5 projection to 2045). I checked the current CDS catalog rather than going from memory, since dataset names/IDs there change.

1. Historical raw weather data (both cities)

- **Dataset:** `reanalysis-era5-single-levels` (ERA5 hourly/daily reanalysis)
- **Variables:** 2m temperature, total precipitation, 10m wind speed, relative humidity
- **Period:** 1990–2020
- **Use:** Baseline for everything — heatwave threshold calculation (Seville), SPI input (Larissa)

2. "Climate indicators for Europe" — worth checking before you pull raw CMIP6 yourself

- **Dataset:** `sis-ecde-climate-indicators` ("Climate indicators for Europe from 1940 to 2100, derived from reanalysis and climate projections")
- **Why this matters:** This is a pre-built product covering both the historical reanalysis period *and* projections out to 2100 in one place, purpose-built for exactly this kind of adaptation-planning question. Check whether it already includes heatwave days and tropical nights for your two cities before you bother pulling raw CMIP6 — if it does, this alone could replace a chunk of your Week 3 work.

3. Raw CMIP6 projections (fallback, if #2 doesn't cover what you need)

- **Dataset:** `projections-cmip6`
- **Model/scenario:** one model (e.g. IPSL-CM6A-LR or MPI-ESM1-2-HR), `ssp5_8_5` experiment
- **Variables:** same as #1
- **Period:** 2025–2045

4. Fire danger — Seville (this is a two-part dataset pair, not one)

- **Historical:** CEMS Fire danger historical dataset (search CDS for "Fire danger indices historical data" — Copernicus Emergency Management Service product)
- **Projections:** "Fire danger indicators for Europe from 1970 to 2098, derived from climate projections" — built on the Canadian Fire Weather Index (FWI) system
- **Important scope note:** this projections dataset uses **RCP4.5/RCP8.5 scenarios**, not SSP5-8.5 — different generation of climate scenarios (CMIP5-era vocabulary, not CMIP6). They're roughly comparable in severity (RCP8.5 $\approx$ high-emissions, same spirit as SSP5-8.5) but aren't the *same* scenario framework. Worth one sentence in your limitations section rather than treating them as interchangeable.

5. Precipitation for Larissa SPI

- Same ERA5 `total_precipitation` variable from #1 covers this — no separate dataset needed, just make sure your Larissa bounding box request includes it
- **Optional cross-check:** E-OBS gridded precipitation dataset, if you want to validate your ERA5-derived SPI against an independent observational source

Summary table

Purpose	Dataset	Citie	Period
Historical baseline (temp, precip, wind, RH)	reanalysis-era5-single-levels	Both	1990–2020
Pre-built indicators (check first)	sis-ecde-climate-	Both	1940–2100
Raw projections (fallback)	projections-cmip6, SSP5-8.5	Both	2025–2045
Fire danger, historical	CEMS Fire danger historical	Seville	1990–2020ish
Fire danger, projected	Fire danger indicators for Europe (RCP-based)	Seville	2025–2045